

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 3

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length = $r\theta$, sector area = $\frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Find the set of values of x for which $2x^2 - 7x + 3 \leq 0$. [3]

2 The equation of a curve is $y = 2x^2 - 8x + 5$.

(a) Express $2x^2 - 8x + 5$ in the form $a(x + b)^2 + c$, where a , b and c are constants. [2]

(b) State the coordinates of the vertex of the curve and the equation of its line of symmetry. [2]

3 A chord AB of a circle with centre O and radius 6 cm subtends an angle of $\frac{2\pi}{3}$ radians at O . Find, in exact form,

(a) the area of the minor segment cut off by AB , [2]

(b) the perimeter of the minor segment. [2]

4

(a) Find the first three terms, in ascending powers of x , in the expansion of $(1 - 2x)^6$. [3]

(b) Hence find the coefficient of x^2 in the expansion of $(1 + 3x)(1 - 2x)^6$. [2]

5 The line l_1 has equation $2x + 3y = 12$. The line l_2 passes through the point $A(5, 5)$ and is perpendicular to l_1 .

(a) Find the equation of l_2 , giving your answer in the form $ax + by = c$. [3]

(b) The lines l_1 and l_2 meet at the point B . Find the coordinates of B and the exact length of AB . [3]

6 The function f is defined by $f(x) = x^2 - 6x + 11$ for $x \geq k$, where k is a constant.

(a) Express $x^2 - 6x + 11$ in the form $(x - a)^2 + b$. [2]

(b) State the smallest value of k for which f has an inverse. [1]

(c) For this value of k , find an expression for $f^{-1}(x)$ and state the domain of f^{-1} . [3]

7

(a) Prove the identity $\frac{\cos \theta}{1 - \sin \theta} - \frac{\cos \theta}{1 + \sin \theta} \equiv 2 \tan \theta$. [3]

(b) Hence solve the equation $\frac{\cos \theta}{1 - \sin \theta} - \frac{\cos \theta}{1 + \sin \theta} = 3 \sin \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [4]

8 A curve has equation $y = 3\sqrt{4x + 1} - 2x$.

(a) Find $\frac{dy}{dx}$. [3]

(b) Find the coordinates of the stationary point of the curve. [2]

(c) Determine the nature of the stationary point. [2]

9

(a) A curve is such that $\frac{dy}{dx} = \frac{12}{(2x+1)^3}$ and the curve passes through the point $(0, 1)$. Find the equation of the curve. [4]

(b) Find the total area of the regions enclosed by the curve $y = x^2 - 2x$, the x -axis and the line $x = 3$. [4]

10

(a) The first term of an arithmetic progression is 12 and the 8th term is 33. The sum of the first n terms is 495. Find the value of n . [4]

(b) A geometric progression has first term 50 and common ratio 0.8.

(i) Find the sum to infinity. [1]

(ii) Find the sum of the terms from the 5th term to the 10th term inclusive. [3]

11 The points $A(-1, 2)$ and $B(5, 10)$ are the ends of a diameter of a circle.

(a) Find the equation of the circle. [3]

(b) Show that the point $C(6, 3)$ lies on the circle, and find the equation of the tangent to the circle at C , giving your answer in the form $ax + by = c$. [3]

(c) The tangent at C meets the x -axis at the point D . Find the coordinates of D and the length of CD . [3]

12 The function f is defined by $f(x) = 2 \cos x - 1$ for $0 \leq x \leq 2\pi$.

(a) State the range of f . [2]

(b) Sketch the graph of $y = f(x)$, showing the coordinates of the points where the graph meets the y -axis and the values of x at the maximum and minimum points. [2]

(c) Describe fully a sequence of two transformations which maps the graph of $y = \cos x$ onto the graph of $y = f(x)$. [2]

(d) Solve the equation $f(x) = 0$, giving your answers in terms of π . [2]

END OF PAPER (Total: 75 marks)